

A PHARMA CLOSEOUT INTELLIGENCE REPORT

AI in Pharma

A Pharma Closeout Strategic Briefing

Five insights every pharma executive needs before their next board meeting. A data-driven analysis of where AI creates value, where it doesn't, and what separates leaders from followers.

Based on synthesis of 60+ analyst reports from BCG, McKinsey, Deloitte, EY, Accenture, S&P Global, Bain, Gartner, DT Consulting, Benchling Research, and others · 15 pharma companies analyzed · 3 in-depth case studies · 6,020 HCP survey responses across 13 countries · Full source index available in the Compendium, Appendix D

EDITION 1
March 2026

The \$275B question: can pharma close the AI execution gap before the patent cliff hits?

01

The largest AI opportunity is commercial (\$18-30B/yr) but it gets only 5-15% of the budget

R&D absorbs 60-80% of AI spend despite lower ROI per dollar. Investment is strategically backwards.

02

75% of AI pilots never reach production — the barrier is organizational, not technical

Five systematic failure modes (governance, data, skills, workflow, change mgmt) compound when addressed sequentially.

03

Leaders share one pattern: they invest in the weakest link, not the most exciting one

Pfizer, UCB, and Novo Nordisk build proprietary platforms, cross-functional governance, and commercial-first sequencing.

\$275B

Revenue at risk from patent cliff by 2030

0%

Pharma TSR over 5 years vs. S&P's 16%

89%

Scientists now use AI as first research stop

SECTION ONE

The Crisis

Why the current pharmaceutical business model is breaking

\$275 billion in annual revenue is losing patent exclusivity by 2030, threatening the blockbuster model's viability

This is not a distant threat. The top 15 companies face a \$240B growth gap by 2030. The replacement pipeline cannot generate returns fast enough through organic innovation alone. Even the most productive R&D organizations cannot offset this erosion within the compressed timeframes imposed by cliffs already underway.

BCG, 'Biopharma Patent Cliff Analysis,' 2025 · EY, 'Growth Gap Analysis,' 2025

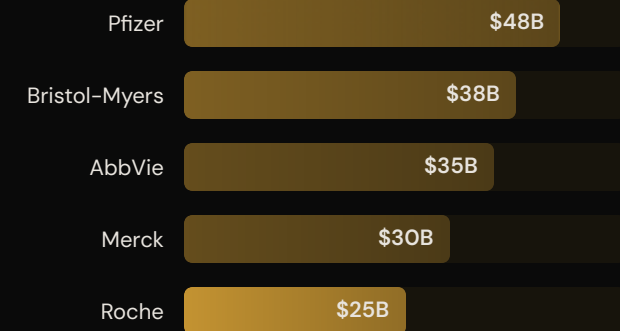
The patent cliff isn't a planning exercise — it's a forcing function. Companies that haven't started AI-driven commercial transformation by 2027 won't close the gap.

\$275B

ANNUAL REVENUE LOSING EXCLUSIVITY 2025–2030

vs. \$180B in the previous 5-year cycle

TOP EXPOSURE BY COMPANY (ILLUSTRATIVE)



80% concentrated in top 20 companies

0%

PHARMA AVERAGE TOTAL SHAREHOLDER RETURN, 2021-2025

**The S&P 500 returned 16% over the same period.
Only 6 of the top 20 pharma companies
outperformed the index.**

This is not a cyclical downturn. It reflects structural economics — the cost of drug development rises while commercial execution and digital engagement deteriorate. The largest, most profitable healthcare sector has destroyed relative shareholder value for half a decade.

This is a crisis of value creation, not scientific productivity. The fix is commercial execution and AI-driven efficiency — not more R&D spending.

BCG (2026) Total Shareholder Return Analysis

Digital spending up 51%, customer experience down 7% – pharma expanded reach without expanding quality

Digital channels now represent 51% of all HCP engagement for the first time. Yet digital CXQ trails non-digital by 5 points (52 vs. 57). Overall customer experience quality declined from 58 to 54 in a single year. The paradox: massive digital investment produced worse outcomes because channels were managed with less governance than traditional engagement.

DT Consulting (2025) · 6,020 HCPs across 13 countries · 12,040 rated experiences

Digital transformation without AI-powered quality governance destroys the customer experience it was meant to improve.

51%

DIGITAL SHARE OF HCP
ENGAGEMENT

Up from 35% in 2022

54

OVERALL CXQ SCORE
(2025)

Down from 58 in 2024

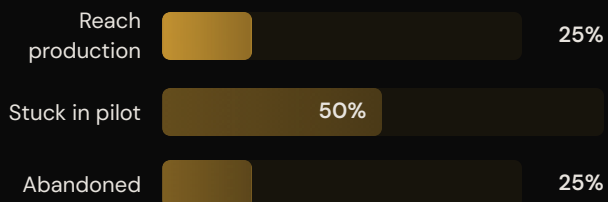
-17pts

ADVERSE EVENT INFO
QUALITY

-15pts

NEW INDICATION INFO
QUALITY

AI PILOT-TO-PRODUCTION TRANSITION



Failure is organizational, not technical

FIVE FAILURE MODES — PREVIEW

Governance, data, skills, workflow, and change management must be addressed in parallel. Sequential capability building compounds delays — see Section Two for the full barrier analysis with prevalence data.

75% of AI pilots never reach production — and the fix isn't better technology

The industry built AI labs with 5–10 data scientists isolated from operations. Pilots consumed 6–9 months and \$1–2M in consulting fees, producing dashboards nobody used. The gap between pilot and production isn't technical — it's organizational. Companies that address all five barriers in parallel succeed; those that work sequentially stall.

Deloitte, 'AI Pilot-to-Production Gap,' 2025 · McKinsey, 'AI Adoption Patterns in Biopharma,' 2025

Stop funding sequential capability building. The five barriers (governance, data, skills, workflow, change) must be addressed in parallel or they compound.

SECTION TWO

Where AI Actually Works

Separating proven value from expensive experiments

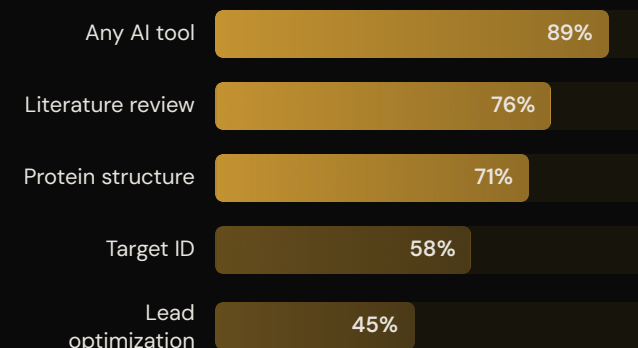
89% of scientists already use AI as their first step – adoption has moved from experimental to mainstream

Benchling survey data: 50% report faster time-to-target, 42% improved hit rates in screening, 56% expect cost reductions within two years. 80% of organizations plan to increase AI budgets, with 23% expecting to double or more. This is conviction about ROI, not cautious optimism.

Benchling Research (2026) · McKinsey GenAI in Biopharma (2025)

Discovery AI adoption is no longer the strategic question. The question is whether the rest of the value chain can absorb what discovery produces.

AI ADOPTION BY DISCOVERY FUNCTION



80%

PLAN TO INCREASE AI
BUDGETS

50%

REPORT FASTER TIME-
TO-TARGET

Manufacturing AI reduces process optimization cycle time by 20-40% and corrective maintenance by 50% – these are production-verified, not pilot-stage

Production supervisors spend 40% of time on manual reporting and information collation. GenAI copilots compress this dramatically. A manufacturing org deploying AI across 30+ processes captures tens of thousands of hours annually. The economics are already proven at scale.

McKinsey (2025) GenAI in Biopharma Operations · BCG (2024) Digital Twins

Manufacturing AI has the best ROI in pharma (3–4x per dollar) yet receives the least attention from boards focused on discovery (Bain & Company, 2025).

20-40%

Reduction in technical solution identification time

50%

Reduction in corrective maintenance workload

5%

Reduction in equipment breakdown time

ROI PER INVESTED DOLLAR

3–4x

Highest across all pharma AI domains (Bain & Company, 2025)

\$18-30B

ANNUAL COMMERCIAL AI OPPORTUNITY — THE LARGEST AND MOST UNDERFUNDED DOMAIN

Commercial represents the single largest GenAI opportunity, larger than R&D (\$15-28B) and clinical (\$13-25B) combined. Yet it receives only 5-15% of pharma's AI budget.

An AI-accelerated molecule that reaches market 18 months faster is worthless if the commercial team's launch playbook is still manual, content generation is pre-AI, and field force lacks next-best-action intelligence. This opportunity gap is pharma's biggest strategic blind spot.

If your commercial AI budget is below 25% of total AI spend, you are systematically underinvesting in the highest-value domain. Rebalance within the existing budget envelope.

McKinsey, 'GenAI in Biopharma,' 2025 · 63 use cases modeled

Companies fail at AI scale due to five systematic organizational barriers — not technology limitations

Companies have access to the same LLMs and platforms as leaders. The gap is organizational. Sequential capability building (governance → skills → data → workflows) creates compound delays. Parallel investment with explicit integration cadence is the only path that works.

Deloitte (2025) · BCG (2025) Commercial AI Deployment

The five barriers are not independent — they interact. Fixing governance without fixing workflow design produces compliant processes nobody uses.

FIVE FAILURE MODES — % OF COMPANIES AFFECTED

Change mgmt 70%

Governance gap 60%

Workflow design 55%

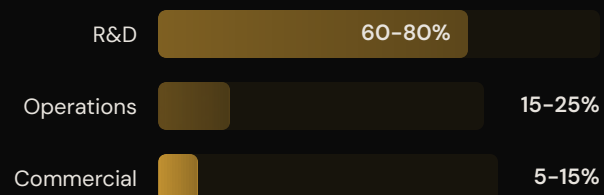
Data gaps 45%

Skill shortage 38%

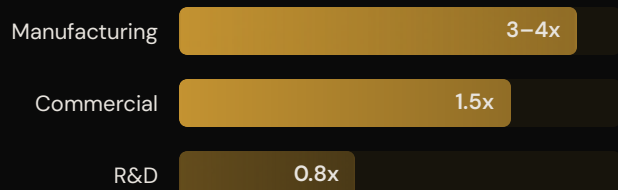
All five must be addressed in parallel

AI BUDGET ALLOCATION VS. ROI PER DOLLAR

BUDGET SHARE



ROI PER DOLLAR



Budget and ROI are inversely correlated

60-80% of AI budget goes to R&D, yet the highest ROI comes from manufacturing and commercial

R&D receives dominant investment because it's visible and scientifically exciting. Manufacturing and commercial deliver superior near-term ROI but get far less. Internal LLM platforms cost 5-30x less per user than enterprise Copilot licenses because they serve targeted use cases.

McKinsey, 'AI Investment Analysis in Pharma,' 2025 · BCG, 'Platform Economics in Biopharma,' 2025

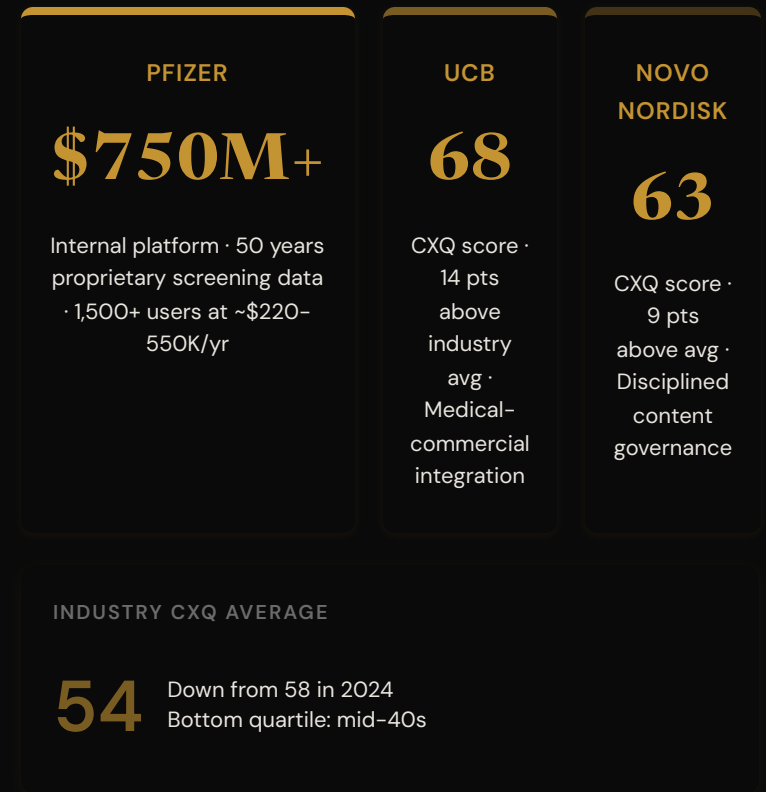
This allocation is strategically backwards. Budget follows excitement rather than return. Manufacturing at 3-4x ROI deserves more than 15-25% of the budget (Bain & Company, 2025).

Industry leaders share three patterns: proprietary data, cross-functional governance, commercial-first sequencing

Leaders built defensible platforms by combining proprietary data with LLMs, creating advantage external vendors cannot replicate. They implemented cross-functional governance where commercial content requires medical review before deployment. They targeted AI at the lowest-scoring customer touchpoints first.

BCG (2025) · DT Consulting (2025) CXQ Benchmarks · EY (2025)

The leader playbook is replicable. UCB built its CXQ advantage at mid-cap scale (\$5B revenue). It's about discipline, not resources.



SECTION THREE

The Cascade Framework

A system-level model for diagnosing and fixing AI value leakage

AI value doesn't accumulate domain by domain – it cascades across four primary links and two substrates where the weakest constrains everything

Looking at domains in isolation conceals a critical insight: acceleration in one domain creates pressure on adjacent domains. Discovery AI that compresses preclinical timelines by 30–50% pushes more compounds into development. If clinical infrastructure is still Level 1, the pipeline backs up. Discovery investment generates only 30–40% of theoretical value.

Compendium proprietary analysis · Cross-domain synthesis, 60 analyst reports

MATURITY LEVELS

- L1 Experimental:** Pilot-stage, isolated use cases, no cross-functional integration.
- L2 Operational:** Deployed in production workflows with measurable KPIs.
- L3 Integrated:** Cross-domain data flows; AI outputs inform adjacent functions.
- L4 Autonomous:** Self-optimizing systems with closed-loop feedback across the value chain.

System-level ROI is constrained by the weakest link. Optimizing the strongest link while starving the weakest is economically irrational.

INDUSTRY AVERAGE CASCADE PROFILE



Commercial is the bottleneck at L1.2

L1.2

INDUSTRY AVERAGE: COMMERCIALIZE LINK MATURITY — THE SYSTEMIC BOTTLENECK

40% of the industry is stuck at Level 1 on commercial AI. Value created upstream in discovery only materializes if commercial can deploy it.

A discovery platform identifying three novel compounds per year is worthless if the commercialize link cannot launch faster than the current 3–4 year cycle. AI-optimized manufacturing that reduces cycle time by 20–40% delivers no value if pricing power has eroded from weak customer engagement.

System-level value maximization requires investing in the weakest link first, not the most exciting one. For 79% of pharma companies, that means commercial.

Compendium self-assessment synthesis · DT Consulting (2025)

Every pharma company exhibits one of five diagnostic patterns — identification determines where to invest first

Pattern identification is not academic. It determines which domain receives the next marginal dollar for maximum system return. The most common pattern (C: Commercial Blind Spot) affects 79% of pharma companies.

Compendium diagnostic framework · Deloitte (2025) · BCG (2025)

Run the self-assessment with your cross-functional leadership team (90 minutes). Shared diagnosis becomes the foundation for aligned execution.

A: PIPELINE BACKUP

Strong Discover (L3+), weak Develop (L1). AI finds more targets than clinical can absorb. Fix: invest in trial modernization.

B: GOVERNANCE CEILING

Multiple domains L2+, governance stuck L1. Board fear of regulatory risk prevents deployment. Fix: governance framework first.

C: COMMERCIAL BLIND SPOT

Most common (79%).
R&D at L2+,
Commercial at L1. Fix:
redirect 25–40% of AI
budget to commercial.

ALSO:

D: Talent Starvation — 5–10 data scientists with limited deployment org. **E: Isolated Islands** — One link L3+, all others L1.

IF COMMERCIALIZE WEAK (MOST COMMON — 79%)

60% to Commercialize · **20%** Workforce · **20%** Governance

\$22–55M over 18 months · 12–18 months to L2

IF GOVERNANCE WEAK + MULTIPLE LINKS L2+

100% to Governance before scaling

\$5–16M infrastructure · 12–18 months to L2

OTHER SCENARIOS:

Discover Strong + Develop Weak: 70% to Develop. **Workforce Constraint:** Hire 30–50% more AI-ready talent.

Strategic budget allocation follows system logic, not domain enthusiasm

The common error: companies optimize their strongest link while starving their weakest. The business case for each link depends on adjacent link maturity. Investing \$55M in discovery AI when commercialize is L1 is economically irrational — it generates only 30–40% of theoretical value due to the downstream bottleneck.

Compendium budget modeling · BCG (2025) · McKinsey (2025)

Budget allocation logic flows from Cascade diagnosis. If you haven't diagnosed your pattern, your budget is allocated by politics, not strategy.

Mid-cap pharma identified Pattern C, redirected \$33M to commercial – CXQ improved 8 points in 18 months

Starting profile: Discover L2.8, Develop L2.1, Deliver L1.9, Commercialize L1.2. Diagnosis: Pattern C. Budget reallocation from 70/20/10 (R&D/Ops/Commercial) to 50/20/30. Interventions: next-best-action engine (\$13M), AI medical information platform (\$9M), governance framework (\$5.5M), talent (\$5.5M). Results: CXQ 51→59, rep adoption 92%, launch market share 15% higher vs. internal forecast.

Compendium case study (anonymized) · Client confidential performance data

The constraint was not molecules — it was market access.
Budget reallocation followed Cascade logic: invest in the weakest link first.

BEFORE (BASELINE)

51

CXQ Score

70% R&D

20% Ops

10% Commercial

AFTER (18 MONTHS)

59

CXQ Score

50% R&D

20% Ops

30% Commercial

PRE-INTERVENTION PROFILE



SECTION FOUR

What Leaders Do Differently

Three case studies in building competitive AI advantage

Pfizer's \$750M-\$1B internal platform creates a data moat that external vendors cannot replicate

Pfizer built an infrastructure layer, not a single AI agent: a unified knowledge graph spanning 50 years of screening data, binding affinities, ADME properties, clinical outcomes, and failed program learnings. Cost: ~\$220-550K annually for 1,500-2,000 users vs. \$5-11M for enterprise Copilot licenses. The knowledge graph grounds LLMs in internal evidence, not hallucinations.

BCG (2025) · Pfizer public statements · Investor relations

Proprietary data is defensibility. Your historical screening data, customer records, and operational metrics are competitive assets. Build integration layers, not vendor dependencies.

INVESTMENT

\$750M-\$1B

Internal platform · 50 years proprietary screening & compound data

COST ADVANTAGE

~\$220-550K

Annual cost for 1,500-2,000 users vs. \$5-11M enterprise Copilot

THREE LESSONS

1. Build, don't buy — resist vendor lock-in
2. Target high-volume, rules-based decisions
3. Build five capabilities in parallel

Recursion acquired Exscientia for \$712M – the largest AI pharma M&A deal signals what traditional pharma will pay for

The value was not technology combination but data integration — combining Recursion's petabytes of cellular imaging (irreplaceable) with Exscientia's AI molecular design (AI compounds already in Phase II). The merged entity creates a closed-loop system where each experiment validates and retrains the AI. This data flywheel creates compounding advantage.

EY (2025) AI M&A · Recursion/Exscientia SEC filings · BCG (2025)

Standalone technology is not valuable. Clinical evidence is the exit currency. Pharma pays for defensive capability — they buy to prevent competitors from owning the data moat.

\$712M

LARGEST AI PHARMA M&A DEAL ON RECORD

WHAT \$712M BOUGHT

Proprietary experimental data (cellular imaging) + Clinical validation (Phase II) + Data flywheel (compounding advantage)

CASCADE GAP

Strong Discover + Deliver loop. But weak Develop and Commercialize — the same challenge facing all pharma.

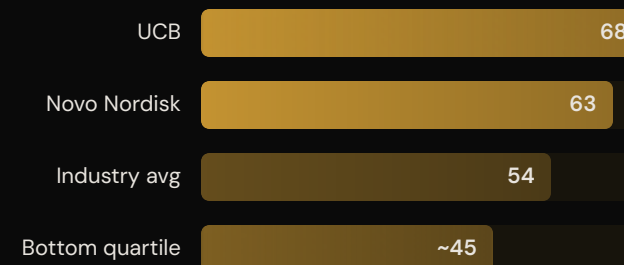
While industry CXQ dropped 7% to 54, UCB maintained 68 and Novo Nordisk 63 – a 14-point premium that translates to market share

CXQ drives approximately one-third of HCP prescription decisions. UCB's 14-point premium translates to measurable market share advantage. Both companies share operational patterns: medical sign-off before commercial deployment, single commercial org per therapeutic area, CXQ as a leadership KPI, and digital held to the same review standards as traditional materials. Crucially, UCB built this advantage at \$5B mid-cap scale.

DT Consulting (2025) · 6,020 HCPs · IPSOS Global Trends (2024)

Closing one maturity level on Commercialize (L1→L2) could yield 5–10 CXQ points. At 33% influence on prescribing, that's quantifiable revenue impact.

CXQ SCORES: LEADERS VS. INDUSTRY



Medical-commercial integration is the differentiator

UCB CASCADE PROFILE (ESTIMATED)



Strong Commercial = competitive moat

SECTION FIVE

The Road Ahead

Three imperatives and a 180-day action framework

Three non-negotiable shifts for 2026-2030: rebalance budget, build governance, establish regulatory precedent

These are not aspirational goals. They are the minimum requirements for companies that want to survive the patent cliff with competitive AI capability. The first movers on regulatory precedent will capture 18–24 month advantages that followers cannot compress.

Deloitte (2025) · BCG (2025) · McKinsey (2025) · Compendium synthesis

The window for action is closing. Companies that haven't started by 2027 will face a gap they cannot close through organic effort alone.

01

Rebalance AI Budget Toward Commercial

Now: 60–80% R&D, 5–15% Commercial

Target by 2027: 45–55% R&D, 25–30% Commercial

02

Board-Level AI Governance by Q4 2026

Now: 66% of boards have no AI risk oversight (Deloitte, 2025)

Target: CAI Officer + quarterly board review + annual assessment

03

File AI-Assisted Regulatory Submission by 2027

Now: 79% of regulated activities at <40% AI adoption

Target: FDA pre-submission by Q2 2026; filing by 2027

Move from strategy to action in 180 days: diagnose, pilot, and establish governance

This is a structured framework designed for cross-functional leadership teams. Each phase builds on the previous one. The Cascade Self-Assessment (Days 1–7) creates shared diagnosis that becomes the foundation for all subsequent investment decisions. Do not skip the assessment — it is more valuable than 10 strategy consultants debating in isolation.

Compendium Appendix B · Full 30-question scorecard in the Compendium

The 180-day framework is not optional. It's the minimum viable action plan. Companies that don't diagnose their Cascade pattern will allocate budget by politics, not strategy.

DAYS 1–30: ASSESS & ALIGN

Complete Cascade Assessment + Baseline CXQ

Convene 7 cross-functional leaders · Administer 6-link diagnostic · Identify pattern (A–E) · Benchmark CXQ against industry avg (54) and leaders (68)

DAYS 31–90: PILOT & GOVERN

Launch Commercial AI Pilot + Governance Framework

Select lowest-scoring CXQ touchpoint · Deploy AI pilot targeting it · Draft AI governance charter · Establish medical-commercial review workflow · Present Cascade diagnosis to board

DAYS 91–180: SCALE & MEASURE

Budget Reallocation + First Measurable Results

Propose AI budget reallocation · Set quarterly governance review cadence · Identify regulatory submission opportunity · Measure CXQ improvement against baseline

CASCADE SELF-ASSESSMENT (SAMPLE QUESTIONS)

DISCOVER

How many AI-assisted compounds have entered human trials?

DEVELOP

What % of clinical protocols leverage AI patient recruitment?

DELIVER

What % of manufacturing includes AI predictive maintenance?

COMMERCIALIZE

What % of reps use AI next-best-action in territory planning?

GOVERNANCE

Does the board have explicit AI risk oversight with quarterly cadence?

Complete the Cascade Self-Assessment in 90 minutes with your leadership team

The process: convene seven leaders (CFO, CTO, Head R&D, Head Manufacturing, CCO, CHRO, Chief Compliance). For each of the four primary links and two substrates, answer five diagnostic questions. Score 1-4. Average per link. Plot against the five patterns. Map to investment sequence. This creates shared diagnosis that eliminates political budget allocation.

Full 30-question scorecard available in the Compendium (Appendix B)

The cross-functional team that completes this assessment together creates shared diagnosis. That shared diagnosis is more valuable than the assessment itself.

How to use this briefing: a reading guide by role

This briefing is designed for six executive personas. Each has a different path through the material based on their role and immediate priorities. The Compendium provides the full analytical backing, case studies, and implementation playbooks referenced throughout.

VP Commercial Excellence / Digital

Start: pp. 9–11 (commercial opportunity) | Then: pp. 13–17 (Cascade framework) | Action: 180-day plan (p. 22)

Strategy Director

Start: pp. 3–6 (the crisis) | Then: pp. 18–20 (case studies) | Action: Self-assessment (p. 23)

Head of BD / M&A

Start: p. 19 (Recursion deal) | Then: pp. 11–12 (investment logic) | Action: Cascade valuation lens

CSO / R&D Leader

Start: pp. 7–8 (where AI works) | Then: pp. 13–14 (cascade bottleneck) | Action: Cross-domain pilot plan

VP / Dir Commercial

Start: pp. 5–6 (CXQ crisis) | Then: pp. 9–11 (commercial opportunity) | Action: Budget reallocation (p. 15)

CFO / Finance Leader

Start: p. 11 (budget vs. ROI) | Then: pp. 15–16 (investment logic) | Action: Cascade ROI modeling

WHAT THIS BRIEFING COVERED

- ✓ The \$150B AI opportunity and pharma's structural underinvestment
- ✓ Commercial AI as the systemic bottleneck (L1.2 industry average)
- ✓ The Cascade Framework: four links, two substrates, five diagnostic patterns
- ✓ Three case studies: Pfizer, Recursion-Exscientia, UCB & Novo Nordisk
- ✓ 180-day action framework and 30-question self-assessment preview

Go Deeper: The Definitive Compendium

This briefing is the condensed strategic layer. The full Compendium provides the analytical depth your team needs to execute — 65,000+ words of frameworks, evidence, and playbooks that go far beyond what any briefing can contain.

[Request the Full Compendium](#)

thepharmacloseout.com

THE COMPENDIUM DELIVERS BEYOND THIS BRIEFING



63 Use Cases with Maturity Assessments

The complete breakdown across all four Cascade links — discovery through commercialization (Appendix H)



Full 30-Question Self-Assessment

Cascade diagnostic scorecard with scoring rubric and pattern-matching guide (Appendix B)



180-Day Action Framework

Phase-gated implementation playbook with budget allocation models and governance templates



Three Full Case Studies

Pfizer, Recursion-Exscientia, UCB & Novo Nordisk — complete analysis with financials and lessons



60 Source Citations

Every claim traced to BCG, McKinsey, Deloitte, EY, DT Consulting originals (Appendix D)

CONSULTING & ADVISORY

Cascade Workshop — Half-day facilitated self-assessment with your leadership team | **Advisory** — AI portfolio investment and execution sequencing | **Custom Reports** — Bespoke competitive intelligence

Daily Intelligence: The Pharma Closeout Podcast (Spotify & Apple Podcasts) · Newsletter (thepharmacloseout.substack.com)